



A question-answering framework for geospatial data retrieval enhanced by a knowledge graph and large language models

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ABSTRACT

The rapid advancement of Earth observation technologies has resulted in an explosive growth of geoscientific data. Efficient retrieval of such data is essential but often challenging, especially for non-experts, due to decentralized data sources and limited semantic support in existing portals. To address these challenges, we propose the Geospatial Data service-oriented Question-Answering (GDQA) system, which combines a geospatial data knowledge graph (GDKG) with large language models (LLMs) to enable natural language-based data discovery. Our framework includes two key components. First, we construct the GDKG by aggregating data from five major portals and employing an information extraction model to automatically identify and structure fine-grained knowledge from unstructured texts. Second, we develop a method called Spatio-Temporal Reasoning on Knowledge Graphs (STRKG), which enables LLMs to perform reasoning and exploration over GDKG, enhancing their ability to understand and respond to complex geoscientific queries. While LLMs contribute to flexible natural language interaction, the GDKG complements them with domain-specific knowledge. To evaluate our system, we design a set of competency questions spanning four dimensions: constraint quantity, reasoning depth, spatiotemporal logic, and inference complexity. Compared to baseline approaches including KGQArr, Text2Cypher, KAPING, and ToG, our method achieves superior performance in both accuracy and efficiency, significantly reducing time and token costs.

ARTICLE HISTORY

Received 6 August 2024
Accepted 17 May 2025

KEYWORDS

Knowledge graph; large language models; geospatial data; question-answering system

1. Introduction

The advancement of Earth observation technology and data processing methods has led to explosive growth in geospatial data related to Earth science. Every day, tens of

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thousands of geospatial data points are transmitted, shared, and utilized worldwide, playing a crucial role in advancing our understanding of the human-environment relationship (Gettelman et al. 2022; Guo et al. 2020; Ortiz Vivar et al. 2022). To promote comprehensive sharing and value-added utilization of geospatial data, countries and organizations around the globe have launched a series of Earth science data contribution programs and established numerous Spatial Data Infrastructures (SDIs) to archive and provide these datasets online (Hu et al. 2015; Wang et al. 2021). For example, China's National Earth System Science Data Center covers five major domains, aggregating a total of 2.4 PB of Earth science data resources. The Earth Engine Data Catalog hosts a vast collection of global and regional datasets, officially provided by Google Earth Engine (GEE). The academic journal Earth System Science Data facilitates the publication and dissemination of papers and data resources related to Earth science. In Earth science research, acquiring the necessary data is often the primary task. For most users of geographic spatial models, preparing large volumes of input data is a time-consuming, labor-intensive, and costly task (Zhu 2019). Therefore, effective searching and discovery of information from these spatial data infrastructures have become increasingly important for researchers. This helps save time and costs related to data preparation, allowing users to focus more on analyzing model computation results.

However, there are two unavoidable challenges when users try to retrieve the data they need. On the one hand, despite substantial efforts focused on SDIs (Wiemann and Bernard 2016), data often remain scattered across the web in isolated silos that are not effectively organized or interconnected. This fragmentation often leads to incomplete access to geoscientific data and inefficient data exchange, thus impeding effective data retrieval and discovery. On the other hand, an increasing number of end-users originate from non-geoscientific disciplines, which impacts the development of user-centric technologies. Most geospatial data portals still rely on the most basic and direct method of keyword matching for data retrieval and discovery services. This simplistic approach can create significant obstacles for non-expert users, who often lack interdisciplinary expertise. As a result, they may need to spend considerable time gathering and filtering through the required data. Integrating multi-source data from different fields is particularly challenging and can be highly demanding (Sudmanns et al. 2019).

To address the aforementioned challenges, we developed a Geospatial Data Knowledge Graph (GDKG) enhanced generative question-answering system to improve data retrieval and data discovery. Our goal is to break down data silos and provide low-barrier data services to both professionals and non-professionals. Our work consists of two core steps: (1) construction of a GDKG; and (2) complementary enhancement of the GDKG and Large Language Models (LLMs) to support generative question-answering services.

In the first phase, we aggregate geospatial data from multiple portals, including both structured and unstructured textual data. For structured data, we transform it into knowledge triples through a rule-matching approach. Knowledge triples are fundamental units of knowledge graphs, consisting of a head entity, a tail entity, and the relationship that connects them. For unstructured text, we utilize an information extraction model based on a pre-trained model to extract structured triples. Our constructed knowledge graph facilitates the annotation and interconnection of diverse geospatial data resources.

This approach transforms the current paradigm, where individual literature or data resources are often isolated and scattered across the Internet.

In the second phase, we introduce a multi-stage knowledge graph reasoning approach based on LLMs to support the Geospatial Data service-oriented Question-Answering system (GDQA). This technology leverages LLMs to perform spatiotemporal exploration and reasoning on the knowledge graph, aiming to retrieve knowledge triples relevant to the user's questions. LLMs, with their exceptional semantic understanding and language generation capabilities, are employed for user intent extraction, knowledge graph reasoning, and ultimately generating responses. The knowledge graph serves as a highly reliable knowledge base, helping to mitigate the hallucination issues inherent in generative artificial intelligence models. GDQA eliminates the need for users to learn knowledge graph query languages, allowing them to interact, retrieve, query, and navigate geospatial data using natural language.

The contributions of this study can be summarized as follows:

1. We construct a geospatial data knowledge graph that integrates knowledge from data platforms of four different organizations and one authoritative academic journal. This knowledge graph not only associates related geospatial data but also automatically extracts fine-grained implicit knowledge from the academic literature.
2. We introduce a novel method named spatiotemporal reasoning on knowledge graph, which utilizes LLMs to interactively explore relevant knowledge on the knowledge graph.
3. We evaluate the precision, recall, time cost, and token cost of GDQA across various dimensions of competency questions. Our method achieves optimal performance and significantly reduces both time cost and token cost compared to existing approaches.

2. Related work

2.1. Advances in LLMs for the GIS domain

In November 2022, OpenAI launched ChatGPT, an intelligent question-answering system based on LLMs, signaling the start of an era dominated by large-scale models built on parametric knowledge (Wu et al. 2023). These LLM-based systems can generate fluent human-like responses for virtually all natural language downstream tasks (Liu et al. 2023; Shen et al. 2023; Zhao et al. 2023). In the GIS domain, researchers have tested the capabilities of LLMs for various GIS-related tasks. Mai et al. (2023) demonstrated that, for GIS tasks involving only text, LLMs can outperform task-specific supervised models. However, while LLMs excel at text-based tasks, relying solely on them for more complex tasks in the GIS domain remains challenging (Wang et al. 2024b). These challenges primarily stem from the capability and domain knowledge boundaries of LLMs in GIS tasks. Capability boundaries refer to tasks that exceed the text generation capabilities of LLMs. Some researchers have begun exploring the use of domain tools or code to enhance large models and construct autonomous agents to expand the boundaries of GIS capabilities. LLM-Geo (Li and Ning 2023) utilized GPT-4 to develop a prototype system, showcasing its capability to complete spatial analysis tasks through case studies. GeoGPT (Zhang et al. 2024) autonomously collects, processes, and analyzes

geospatial data based on LLMs' robust semantic understanding performance. MapGPT (Fernandez and Dube 2023) combines large language models (LLMs) with spatial data processing techniques to enhance natural language understanding and spatial data analysis. LLM-Find (Ning et al. 2024) provides metadata for data retrieval to LLMs and utilizes LLMs' code generation capabilities to achieve geospatial data retrieval. These studies depend on the inherent knowledge of LLMs as the primary reasoning mechanism to complete tasks, with almost no additional domain knowledge integrated. Therefore, some research efforts have attempted to construct domain knowledge bases to extend the knowledge boundaries of LLMs in the GIS domain. Wang et al. (2024a) utilized LLMs to interact with an industrial park knowledge graph to assist in industrial park planning and operations. Zhu et al. (2024) enabled LLMs to generate accurate information about flood disasters under the constraints of a flood-related knowledge graph. GeoQAMap (Feng, Ding, and Xiao 2023) uses Wikidata as an external knowledge base and transforms text into SPARQL query language through LLMs to implement a geoscience domain question-answering system. However, most of the aforementioned works, such as GeoGPT, MapGPT, LLM-Find, LLM-Geo, and GeoQAMap, have primarily demonstrated their feasibility through case studies and lack quantified experiments to evaluate their performance. In our work, we focus on using the knowledge graph as an external knowledge base to enhance the performance of LLMs in spatiotemporal data retrieval and discovery tasks. In the experimental section, we design detailed evaluation metrics and analyze the results through quantitative experiments and case studies.

2.2. Advances in geographic data retrieval with knowledge graphs and LLMs

In the early stages of geographic data retrieval and discovery, researchers constructed semantic associations and interoperability among datasets through linked data and Semantic Web technologies (Battle and Kolas 2011; Kuhn, Kauppinen, and Janowicz 2014; Sun et al. 2019). For instance, Yue et al. (2016) explored the role of Linked Data in spatial data infrastructures, enhancing the sharing and discovery of geospatial resources. Wu et al. (2022) advanced the cross-domain analysis of heterogeneous climate datasets using climate analysis ontologies and linked data techniques. In addition to Linked Data, the knowledge graph is another important technology within the Semantic Web. Bucher et al. (2020) advocated constructing a knowledge graph about geographic digital assets to improve data access and discovery. Liu et al. (2021) proposed a heterogeneous geospatial data retrieval method based on knowledge graphs, relying on the construction of SQL statements. Although knowledge graph technology significantly enhances the semantic association and interoperability of spatial data, its limitations in natural language understanding and generation capabilities restrict its application in handling more complex user queries (Pan et al. 2024). Therefore, in early semantic enhancement efforts, semantic feature computation was widely used in the retrieval and discovery of geospatial data. Jiang et al. (2018) extracted metadata features from users' query history and proposed a method for discovering and retrieving geospatial data by calculating semantic similarity. Zhang et al. (2021) constructed a knowledge graph using spatiotemporal similarity measures to establish meaningful semantic connections between data products. However, these works, due to a lack of semantic reasoning, led to incomplete and inaccurate search results. With the rapid

development of LLMs and knowledge graph technologies, researchers have recognized the benefits of LLMs in natural language understanding and generation, as well as the strengths of knowledge graphs in constructing, storing, and reasoning about domain knowledge. Integrating these large models and knowledge graphs to enhance the capabilities of domain-specific question-answering systems has become a research trend. Wang et al. (2023) introduced a knowledge-driven thought chain question-answering system that accesses external knowledge and interacts with LLMs. Sun et al. (2024) employed LLMs as agents to interactively explore knowledge-enhanced question-answering systems' knowledge bases. Jiang et al. (2023) explored how LLMs could be used for question-answering tasks on structured knowledge graphs. These efforts primarily focused on solutions based on LLMs for general-domain knowledge graphs. In contrast, our work focuses on geospatial data knowledge graphs, addressing the unique challenges of geospatial data retrieval tasks, including spatiotemporal characteristics, terminological diversity, and inference conditions.

3. Framework

Constructing a knowledge graph and question-answering system for geospatial data is challenging but yields highly valuable outcomes. Our aim is to create a large-scale, multi-source heterogeneous knowledge graph that serves as a foundational knowledge base, facilitating interoperability across different platforms for geospatial data. Furthermore, by leveraging the complementary strengths of LLMs and the knowledge graph, we support the retrieval and discovery of geospatial data for both professional and non-professional users. The proposed GDQA framework is illustrated in Figure 1, where the direction of the arrows represents the flow of the pipeline in the framework. The following sections elaborate on all the GDQA components in detail.

3.1. Data collection and processing

The knowledge graph GDKG constructed in this paper is sourced from five online databases within the geoscience domain, as shown in Figure 2. To facilitate the aggregation of these resources, a web crawler system based on the Scrapy framework was developed. This system is structured around two principal operations: (1) the construction of a pipeline that retrieves the HTML source code from the URLs of all relevant online databases, and (2) the application of XPath expressions and regular expressions to parse these HTML source codes for the extraction of targeted information. The specific online databases used are as follows:

- Earth System Science Data (ESSD).¹ ESSD is an international, interdisciplinary journal for the publication of articles on original research datasets.
- Earth Engine Data Catalog (EEDC).² EEDC contains the official data catalog of Google Earth Engine and includes a variety of standard Earth science raster datasets. As Earth Engine assets, datasets on EEDC are widely shared publicly.
- Climate Data Store.³ Climate Data Store is a data platform of the Copernicus Climate Change Service (C3S), which offers extensive datasets on global and European meteorology and hydrology. It allows free access.

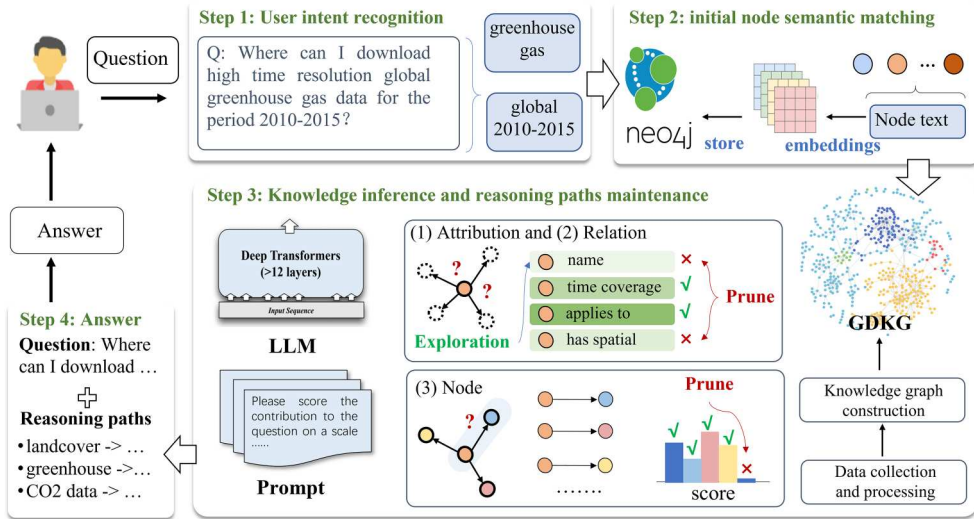


Figure 1. The proposed GDQA framework.

- Awesome Gee Community Catalog (AGCC).⁴ AGCC comprises geospatial datasets sourced from the community and is made publicly available for the broader Google Earth Engine community.
- Resource and Environment Science and Data Center (RESD).⁵ RESD integrates data related to resources and the environment, including remote sensing data, empirical social and economic data, and meteorological data.

3.2. Ontology design

Ontology represents knowledge metadata by defining the entities, attributes, relationships, and constraints of that knowledge. The geospatial data ontology presented in this paper provides a unified formalized representation and semantic foundation for sharing geographic spatial data knowledge. During the ontology design phase, we

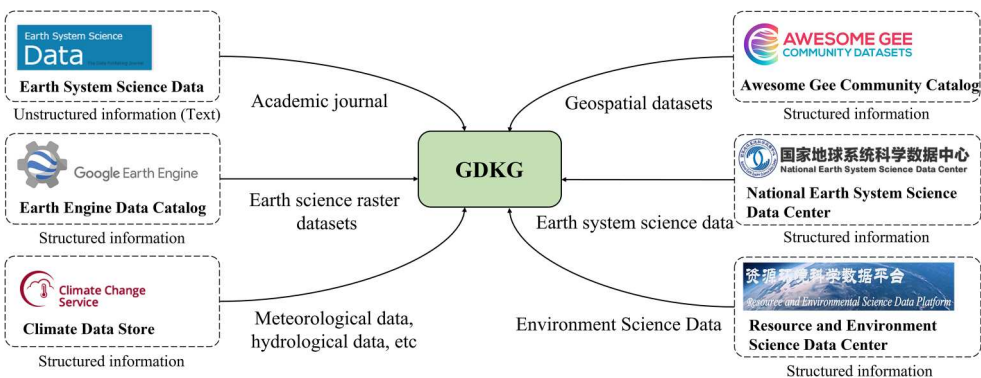


Figure 2. Dataset sources integrated into GDKG.

prioritized reusing GeoDataOnt (Sun et al. 2019), which is highly relevant to the scope of our study. GeoDataOnt defines three essential characteristics of geographic spatial data, specifically addressing the fundamental questions ‘what, where, and when’ about the data – namely, theme, spatial, and temporal coverage (Sun et al. 2019). Additionally, we adopted a bottom-up strategy to enrich and refine additional necessary entities and attributes by cleansing and categorizing the collected data.

Specifically, the ontology developed for GDKG includes the following entities and their associated attributes:

- **Data:** Data are the core of the knowledge graph constructed in this study. The Data entity encompasses a rich set of attributes, including name, temporal coverage, and access link.
- **Paper:** Since a significant amount of geospatial data are presented in academic papers, it is crucial to include papers as an entity type within the ontology. This inclusion makes it easier to access source information about geospatial data within the knowledge graph. Attributes of the Paper entity include the authors, title, and affiliated institutions.
- **Spatial coverage:** Spatial coverage represents the geographical extent of data, which includes geographic coordinates, area description, and geographic name.
- **Sensor:** Sensors are the primary tools for acquiring geospatial data. Within the ontological framework, sensors are abstracted as a key entity type, aiding in understanding the context and origin of data collection.
- **Theme:** Themes are crucial dimensions for categorizing and understanding geospatial data. In the ontology framework, theme entities represent various natural and human-made phenomena that Earth Science data can encompass, such as climate change, geological structures, ecosystems, and human activities. This helps users identify and select datasets relevant to their research interests.
- **Variable:** The variable entity represents various measurable and observable indicators in Earth science data. This includes temperature, humidity, and precipitation in climate data and water temperature, salinity, and flow rate in oceanographic data.
- **Application:** The application entity depicts practical scenarios where Earth Science data are applied to solve real-world problems. Each application entity is linked to specific Earth science data points, revealing the practical value and application potential of the data.

Additionally, we established relationships between entities to represent the knowledge of geospatial data, as shown in Figure 3. For example, ‘*applies_to*’ indicates the relationship between Data and Application, while ‘*has_spatial_coverage*’ denotes the relationship between Data and Spatial coverage. Our ontology provides a comprehensive framework for understanding and exploring Earth science data.

3.3. Construction of the geospatial data knowledge graph

In formal terms, the knowledge graph $GDKG = (N, E)$ is represented as a directed graph that includes a set of nodes N and edges E . The node types correspond to the entity types defined in the ontology and contain various attribute information. The

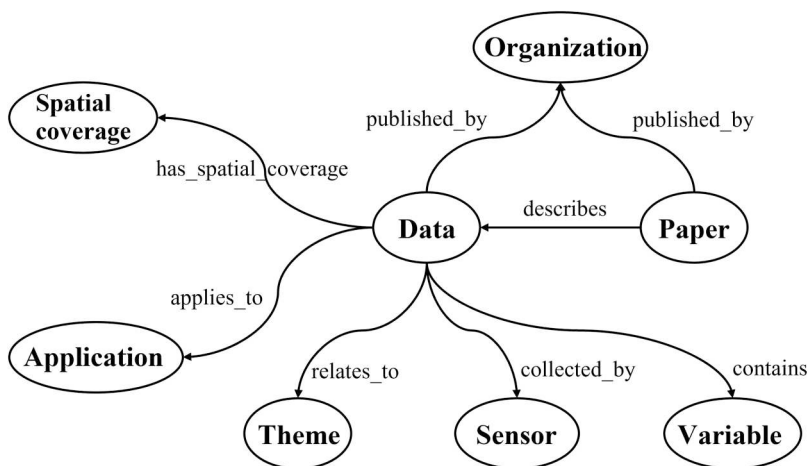


Figure 3. GDKG ontology.

edges illustrate semantic relationships between these nodes. Each node n within the node set N of the knowledge graph possesses a unique identifier. For example, $n.id = 'n.0b24'$, and this identifier can be queried to access the attributes of the node, such as $n.name = 'land surface change dataset.'$

Our knowledge graph, GDKG, is built from both structured and unstructured data, encompassing relevant information and knowledge about geospatial data. Structured data primarily comes from online databases with clearly specified metadata formats. In contrast, unstructured data originates from descriptive texts about geospatial data. Compared to structured data, extracting knowledge from unstructured text requires sophisticated natural language processing techniques. However, unstructured textual documents also contain more professional and rich knowledge, including conclusions and experiences about specific geospatial data articulated by experts in the field. This valuable source of knowledge gives GDKG a unique advantage in supporting downstream domain-specific question-answering tasks. Specifically, for structured data, we construct the knowledge graph by creating a mapping relationship between the crawled data to the ontology. In practice, it is also feasible to consider the attributes of the 'Paper' node, such as authors, as part of the node, facilitating queries about which scholar published a particular dataset. However, given the focus of this research on the retrieval of geospatial data, some additional information (such as the author) is incorporated as attributes of the nodes rather than as separate nodes to simplify the reasoning of the knowledge graph.

For unstructured data, we extract named entities from the abstracts of the literature as instantiated factual knowledge and map these instances to the ontology as nodes or attributes of the knowledge graph. Extracting structured knowledge from unstructured texts in specialized fields typically incurs high costs, particularly due to the lack of annotated text corpora. Our study utilizes the Earth Science Data Corpus (ESDC) from the Science Data Bank,⁶ which includes entity types such as data, spatial coverage, temporal coverage, sensor, theme, variable, and application. ESDC employs three types of entity span classification: flat, nested, and discontinuous entity spans. This fine-

grained annotation ensures high accuracy in semantic tagging but places greater demands on information extraction models. For example, classic information extraction models based on BERT (Devlin et al. 2018) and BiLSTM-CRF (Huang, Xu, and Yu 2015) are not applicable because they cannot recognize nested and discontinuous entity spans. In this study, we utilize the BARTNER model (Yan et al. 2021) to extract knowledge from abstracts of the literature. The BARTNER model uses a unified generative framework to simultaneously solve three different entity spans. During training, the BART-large model⁷ is used as a pre-training model, with a batch size of 8, and an epoch count of 100. Based on this hyperparameter setting, the model achieved an accuracy of 0.613, a recall of 0.675, and an F1 score of 0.642 on the test corpus. Subsequently, we applied the trained knowledge extraction model to the inference stage. The corpus used in this stage consists of the abstracts of all articles in ESSD journals, totaling 1243 as of July 31, 2023.

In the end, GDKG consists of 14830 nodes and 20998 relationships. We store GDKG in the graph database Neo4j. Figure 4 illustrates a fragment of the GDKG. The nodes in the fragment encompass a variety of types, including theme, variable, spatial coverage, data, and application. Each type is distinguished by a unique color. For instance, the node labeled ‘new sea level dataset’ classified as data, displays detailed information in its node properties window, such as ID, description, name, time, and URL. The relationships in the example, such as ‘*applies_to*’, ‘*has_spatial_coverage*’, and ‘*contains*’ interconnect these nodes. Notably, the ‘new sea level dataset’ node is connected to several application-type nodes through the ‘*applies_to*’ relationship, emphasizing the practical applications where this data can be utilized. The GDKG structurally represents geospatial data resources as knowledge triplets, creating associations and organizing data from various sources and structures. This interconnected network of geospatial data enhances their query capabilities, interoperability, and reusability.

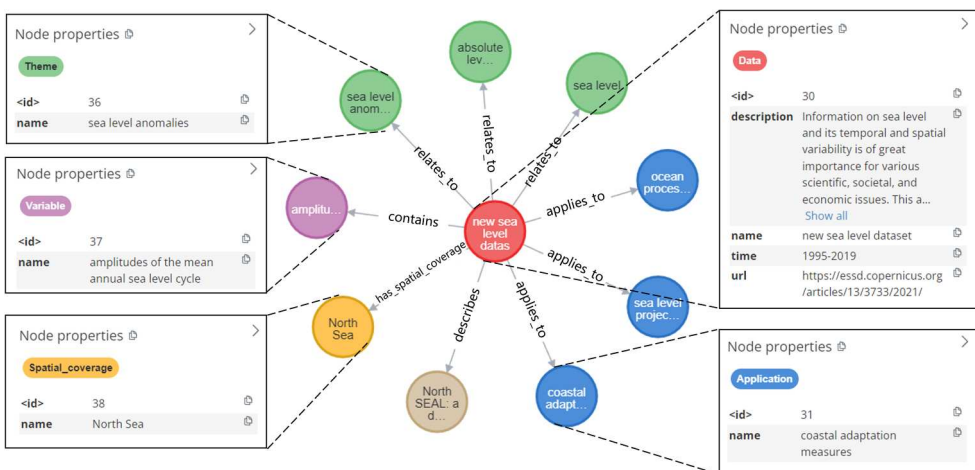


Figure 4. An example of a fragment of the GDKG.

3.4. Spatiotemporal reasoning on the knowledge graph utilizing LLMs

Building on the well-structured GDKG, we developed a geospatial data question-answering system called GDQA. The core strategy of the GDQA is to harness the complementary strengths of LLMs and the knowledge graph. LLMs utilize their superior natural language understanding capabilities to perform spatiotemporal reasoning on the knowledge graph, thereby obtaining accurate knowledge triples essential for solving queries. Spatiotemporal reasoning is defined as the process by which LLMs explore and prune knowledge triples based on temporal and spatial conditions. Concurrently, GDKG provides precise and structured knowledge for knowledge reasoning to alleviate AI hallucinations in specialized domains faced by general LLMs. Thus, LLMs and the GDKG are not separate but work synergistically in the GDQA. In the following sections, we elaborate on the operational and interactive mechanisms between the knowledge graph and the LLMs.

(1) User intent recognition and initial node semantic matching

When a user submits a natural language query Q , GDQA identifies the intent entities from the query and calculates the semantic embeddings for these entities. We use a few-shot learning approach for intent recognition with LLMs, which can extract key information from user queries. Few-shot learning is a technique that significantly enhances the performance of LLMs on specific tasks without the need to retrain the model's parameters. It achieves this by providing the LLMs with a small number of carefully designed examples. The extracted set of intent entities is defined as $E = [e_1, e_2, \dots, e_N]$, where N is constrained to a maximum of three. We propose utilizing the GPT-3.5-Turbo model,⁸ as its capabilities are typically sufficient for this task. The semantic embeddings are calculated using a text embedding model named BAAI General Embedding (BGE).⁹ Meanwhile, we pre-compute the embeddings for all nodes within the GDKG using the same BGE model. The computed embeddings are represented as 256-dimensional dense vectors, which are stored in the vector database Chroma to enhance the efficiency of the semantic matching process. By calculating the semantic similarity metric between intent entities and nodes, we obtain the top K nodes that exhibit similarity scores above 0.8. These nodes are selected as initial nodes for the subsequent knowledge exploration.

(2) Knowledge reasoning and reasoning paths maintenance

The Spatio-Temporal Reasoning on Knowledge Graph (STRKG) utilizes the natural language capabilities of LLMs to conduct extensive multi-hop explorations within the knowledge graph. This exploration continues, with reasoning paths being maintained until the LLM identifies a path that sufficiently supports question answering. The defined format for these reasoning paths is '*entity {attribute} - relationship -> entity {attribute} - relationship -> entity {attribute}*.' Subsequently, we delineate three progressive exploration and pruning strategies, each building upon the previous one, to enhance the effectiveness of knowledge traversal.

- **Attribution exploration and pruning**

During the attribution exploration phase, we obtain a set of all attributes of the initial nodes or tail node in the reasoning path by executing the Neo4j predefined interface. To accurately answer questions and identify relevant attributes, we implement an attribute pruning method based on prompt engineering to determine whether attributes contribute to answering user queries. Prompt engineering is the process of designing and optimizing the inputs for LLMs to enhance the responsiveness of generative artificial intelligence. The set of attributes that contribute to user queries is preserved in the reasoning path. The reasoning path is defined as $P_j = \{(n_d, attr_d^s, value_d^s)\}$, where j is the length of the current reasoning path, n_d is the d -th node, and $attr_d^s$ and $value_d^s$ denote the attribute name and value of the node n_d , respectively. It is important to note that temporal information is included in the attributes of certain nodes. Leveraging the natural language understanding capabilities of large models, our method discriminates time information. If a temporal attribute conflicts with the user's temporal intent, the entire reasoning path associated with the node n_d is removed.

- **Relation exploration and pruning**

We gather all relations connected to the tail node of the current reasoning path as candidate relations. Next, we use relation pruning through prompt engineering to retain only those relations from the candidate set that are valuable for addressing query Q . The reasoning path is updated to $P_j = \{(n_d, attr_d^s, value_d^s), (n_d, ref_d^t)\}$, where ref_d^t represents the t -th relation associated with the node n_d that is preserved. This selective filtering of relations helps in narrowing the scope of the exploration path.

- **Node exploration and pruning**

The initiation of node exploration signifies the progression of the reasoning path to the next hop. In this phase, we sequentially collect the nodes connected by the tail relations within each reasoning path. These neighboring nodes are gathered into the candidate node set N_{cand} . The collection process can be performed through the Neo4j predefined interface. Subsequently, the query Q and N_{cand} are input to the prompt engineering for node pruning, where the top H nodes most relevant to the query Q are selected. The value of H is set to be no greater than 10. This procedure helps prune the potentially vast set of neighboring nodes, filtering out irrelevant noise nodes. Finally, the reasoning path is updated to $P_j = \{(n_d, attr_d^s, value_d^s), (n_d, ref_d^t, n_d^e)\}$, where n_d^e is the e -th first-order neighbor node of node n_d . The maximum number of hops in our reasoning paths is limited to three. Each hop is also referred to as a 'depth' in the knowledge graph exploration.

(3) Evaluation and output from LLMs

After each round of exploration and pruning, the LLMs evaluate whether the current reasoning path sufficiently addresses the user's query. If the evaluation is affirmative, the reasoning paths are used as contextual input for the LLMs to respond to the query.

Additionally, we clarify that the response should be formatted in a way that is easily readable by humans. For example, if multiple suitable data sources are available, the LLMs should organize the search results into a tabular format. If the evaluation is negative, it suggests that the current knowledge graph may not contain result that meets the user's needs. In this case, the GDQA will proceed to deeper knowledge exploration or indicate that relevant knowledge cannot be found. Through this process, the LLMs guide the GDKG to accurately discover knowledge and generate natural language outputs, thereby enhancing the LLMs' responses to domain-specific queries.

4. Experiments

4.1. Experimental design

In this section, we evaluate the GDQA based on various capability dimensions that address user requirements. These dimensions include the number of constraints, the number of hops, spatiotemporal conditions, and inference conditions. We focus on balancing question-answering performance with the associated time and token costs. In the experimental setup, the GPT-3.5-Turbo is utilized for intent recognition as it is generally sufficient. We compare the proposed STRKG method with four other methods: KGQArr (Hu et al. 2023), Text2Cypher, KAPING (Baek, Alham, and Amir 2023), and ToG (Sun et al. 2024). KGQArr utilizes Pre-trained Language Models (PLMs) for similarity calculations to retrieve a set of candidate answers with the highest similarity for the questions posed. KGQArr is a PLM-based method that does not rely on LLMs. The remaining three methods are LLM-based. Text2Cypher uses LLMs to convert user language into Cypher queries, which is a common approach in knowledge graph question-answering tasks. KAPING retrieves additional information from the knowledge graph to enhance LLM prompts without employing any reasoning strategy. ToG implements beam search techniques on the knowledge graph using LLMs. For KGQArr, four different PLMs are provided: BERT (Devlin et al. 2018), RoBERTa (Liu et al. 2019), DistilBERT (Sanh et al. 2020), and Luke (Yamada et al. 2020). For methods based on LLMs, we assess four LLMs: gpt-4-0125-preview¹⁰ (referred to as GPT-4), Qwen2-72b¹¹ (referred to as Qwen2), Baichuan3-Turbo¹² (referred to as Baichuan3), and Llama-3.1-70b¹³ (referred to as Llama3.1). GPT-4, Qwen2, and Baichuan3 are commercial LLMs. Llama 3.1 is an open-source LLM with 70 billion parameters, deployed on four NVIDIA RTX 6000 GPUs (each with 40GB of memory) to ensure comparable performance evaluation. Due to the integrated impact of the selected models (including LLMs and PLMs) and methods on performance, the experimental outcomes are uniquely shaped by the combinations of models and methods used. By analyzing these combinations, we can better understand the strengths and limitations of each integration, guiding optimization strategies for enhanced question-answering performance.

4.2. Competency question design

In our evaluation, we established a set of domain-specific competency questions aligned with user needs to evaluate the capabilities of the GDQA in data retrieval and discovery. This set includes 100 carefully designed questions, covering four dimensions: number of

constraints, number of hops, spatiotemporal conditions, and inference conditions. These dimensions capture the diversity and complexity of user requirements in spatiotemporal data retrieval and discovery. Specifically, the dimensions are as follows:

(1) Number of constraints

The competency questions involve a different number of constraints from the user's intent, ranging from single-constraint queries, such as querying all data for Wuhan, to double-constraint queries, such as querying temperature data for Wuhan, and triple-constraint queries, such as querying temperature data for Wuhan in 2023. The complexity of the competency questions increases with the number of constraints.

(2) Number of hops

The number of hops reflects the number of intermediate knowledge nodes or logical deductions required to answer questions. Multi-hop exploration typically aims to collect more information to create a complete evidence chain for the correct answer. For example, the question 'Which organizations provided data on Arctic topography in 2016?' is a multi-hop issue that necessitates first identifying the available data on Arctic topography in 2016 and then querying the organizations that provided these data.

(3) Spatiotemporal conditions

Non-spatial constraints can be directly matched through semantic computation. Spatiotemporal conditions reflect the user's intent related to both space and time for the task. In our use case scenarios, spatiotemporal matching involves assessing whether the spatiotemporal scope of a dataset aligns with the user's spatial intent. For example, the query 'the temperature in Wuhan in 2023' requires the system to consider not only datasets specifically for Wuhan but also those covering broader areas like China, since Wuhan is included within China.

(4) Inference conditions

These conditions are not explicitly stated in user queries but are essential for generating accurate responses. Answering questions about inference conditions requires a question-answering system with strong semantic understanding and situational reasoning capabilities to capture and interpret the underlying meaning behind user language. For example, a user query such as '*I need the highest temporal resolution temperature data for Wuhan in 2023*' does not specify the exact desired resolution. The system must infer, from the context of the knowledge graph, which dataset meets the user's needs by offering the highest resolution among all Wuhan temperature datasets for 2023.

A competency question typically involves multiple dimensions. Based on their complexity, we categorize these questions into four levels:

- Level 1: Questions that involve only a single constraint and a single hop.
- Level 2: Questions that involve multiple constraints and multiple hops.

- Level 3: Questions that involve spatiotemporal conditions, multiple constraints, and multiple hops.
- Level 4: Questions that incorporate spatiotemporal conditions, multiple constraints, multiple hops, and inference conditions.

Table 1 presents several examples of competency problems, along with their respective levels and the number of competency questions for each of the four levels. Due to the generative nature of LLMs, their output can vary, leading to significant biases when conducting a single test on the test data. To mitigate this, we conduct four evaluations for each competency question and calculate metrics to assess performance across different combinations of models and methods.

4.3. Evaluation method and metrics

Based on the designed competency questions, we manually annotated each question with one or more precise and correct answers. The answers are defined as the ground truth of each question, derived from specific information at one or more nodes in the knowledge graph. The collection and annotation of each question’s ground truth is a labor-intensive and time-consuming task. The specific steps involved are as follows: (1) Initially, we replicated the user intent recognition and initial node semantic matching processes outlined in Section 3.4. This helped to narrow down the range of data for manual annotation; (2) Then, we manually collected fragments of the knowledge graph starting from the initial nodes, which include information about the first, second, and third-order neighbor nodes and the relationships between these nodes. (3) Based on the collected knowledge graph fragments, we manually selected and annotated the ground truth

Table 1. Examples of competency problems by level with corresponding number of questions.

Level	Number	Examples of competency question
Level 1	20	1. Please give me some datasets on the theme of impervious-surface and give me the names of these datasets. 2. I would like some data sets that can be applied to carbon cycle research, please tell me the access links. 3. I want a dataset on the topic of land use or land cover, please tell me the name of the dataset.
Level 2	22	1. I would like to know what are the application scenarios of glacier mass balance and ice sheet mass balance related data sets. 2. I would like some data sets related to the theme of carbon budget that can be applied to carbon cycle research, please tell me the access links 3. I want to know which sensors are used in the dataset with the theme of Land use and land cover.
Level 3	32	1. I want some data sets covering China specifically that can be applied to carbon cycle research. 2. I would like a data set on glacier mass balance covering Switzerland, please tell me the data set name and access link 3. I need global sea surface temperature data products, please tell me the names and access links of these data products
Level 4	26	1. I want the impervious-surface dataset with the highest temporal resolution and global coverage, please tell me the download address of the dataset. 2. I would like a latest dataset related to glacier mass balance and ice sheet mass balance. Please provide me with the dataset name, time range, and access link to the dataset 3. Where can I download the Land use and land cover dataset with the largest time span, please tell me the dataset name, the time range of the dataset, and the access link.

corresponding to the competency questions. Our annotation team consisted of three Master's students and one Ph.D. student, all specializing in geography. During the annotation process, the Master's students, serving as annotators, identified the ground truth for each question by analyzing the collected nodes, attributes, and relationships. The focus was on whether the temporal span and spatial coverage met the users' spatiotemporal requirements. The Ph.D. student served as a reviewer, performing comprehensive consistency checks to ensure the uniformity of the ground truth provided by multiple annotators. (4) Finally, we recorded the competency questions, ground truth, corresponding levels, and other relevant information in a JSON file for efficient batch processing and evaluation. During the evaluation phase, we employed a regularization method to extract all predicted answers from the outputs of the LLMs. These predicted answers were then compared against the ground truth stored in the JSON file to calculate precision and recall.

Our evaluation metrics, including precision, recall, time cost, and token cost, are employed to quantitatively assess the performance of various combinations of models and methods in competency questions. Precision is defined as the ratio of the number of correctly predicted answers to the total number of predicted answers, while recall measures the ratio of the number of correctly predicted answers to the total number of ground truth instances. The formulas for precision and recall are as follows:

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

Where *Precision* and *Recall* correspond to the calculated results of precision and recall, respectively. *TP* (True Positives) represents the number of correctly predicted answers that match the ground truth. *FP* (False Positives) indicates the number of instances where the model incorrectly predicted answers that do not align with the ground truth. *FN* (False Negatives) denotes the number of ground truth instances missed in the model's predicted answers.

Time cost and token cost measure the time spent and the number of tokens consumed by the LLMs. Tokens typically refer to the smallest unit of text processed by LLMs. For commercial LLMs, the higher the token cost, the higher the financial costs incurred. Assessing time cost and token cost is crucial as they directly impact the user's interactive experience and the associated costs.

4.4. Performance evaluation

Table 2 and Table 3 present the precision and recall for various combinations of models and methods applied to the competency questions. We found that the combination of our proposed STRKG method with GPT-4 achieved the best performance. Compared with Qwen2 and Baichuan3, GPT-4 exhibits higher intelligence and stronger reasoning ability. As expected, GPT-4 supports higher performance in Text2Cypher, KAPING, ToG, and STRKG compared to the other two LLMs. Llama 3.1 performs the worst among all the LLMs in the competency question tests, likely due to its relatively lower

Table 2. Performance of different combinations of LLMs and methods.

	GPT-4		Qwen2		Baichuan3		Llama 3.1	
	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall
Text2Cypher	0.262	0.139	0.133	0.053	0.192	0.083	0.134	0.076
KAPING	0.328	0.276	0.287	0.242	0.265	0.227	0.285	0.202
ToG	0.549	0.737	0.539	0.601	0.485	0.582	0.424	0.546
STRKG	0.776	0.795	0.702	0.767	0.548	0.664	0.464	0.603

level of intelligence. When comparing different methods, we found that Text2Cypher had the lowest performance. This is because Text2Cypher directly uses segments of the original text from the query as part of the Cypher query, lacking the capability for semantic fuzzy querying. Thus, Text2Cypher fails when faced with different languages or varied terminological expressions. The performance of KGQArr, which is based on PLMs, was equally unsatisfactory, particularly in terms of precision. This may be due to the PLMs not being further trained on geoscience-specific datasets, resulting in inadequate performance in understanding and applying specialized geoscientific knowledge. However, despite the absence of domain-specific training data, our proposed STRKG method combined with GPT-4 was able to achieve nearly 0.8 in both precision and recall. This highlights the benefits of LLM-based question-answering methods in quickly launching new domain-specific QA initiatives without requiring extensive further training.

Subsequently, we evaluated the performance of different combinations of models and methods on competency questions across four levels, as shown in Figure 5. In Figure 5, line styles differentiate between precision and recall, while line colors represent different methods. The specific models employed by each method are labeled above their respective line charts. Text2Cypher achieves relatively good precision on the less challenging Level 1 competency questions; however, its performance significantly declines on the more complex Levels 2, 3, and 4. KAPING exhibits a similar trend, excelling only at the most basic Level 1 competency questions. In contrast, our proposed STRKG method shows significant advantages in both precision and recall across all four levels. Compared to Levels 2 and 3, both Text2Cypher and KAPING exhibit minor improvements in handling Level 4 competency questions. However, neither method is fundamentally capable of understanding and processing questions that involve inference conditions. We attribute this limitation to the artificial intelligence hallucination associated with LLMs, which results in a small percentage of correct answers being guessed accurately. KGQArr exhibits poor performance across all four levels of competency questions. Its reliance on similarity matching to determine answers fails to capture the full semantics of the questions, particularly when they involve multiple constraints and spatiotemporal conditions that represent user intent. Furthermore, KGQArr lacks the capability to utilize the structured knowledge within knowledge

Table 3. Performance of different combinations of PLMs and method.

	Bert		Roberta		DistilBert		Luke	
	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall
KGQArr	0.191	0.270	0.172	0.284	0.142	0.280	0.178	0.294

graphs for reasoning, which limits its effectiveness in addressing the more complex questions at Levels 3 and 4.

In addition to assessing the performance of the answers, we also considered the time cost and token cost associated with different models and methods. Figure 6 illustrates the relationship between recall and costs for various combinations of models and methods. In Figure 6 (a), which compares recall and time cost, the horizontal axis represents recall, while the vertical axis represents time cost. A position closer to the bottom right corner indicates higher recall and lower time cost. From the plot, we can see that STRKG, using Qwen2 and GPT-4, achieved relatively high recall but also incurred significant time costs, especially GPT-4. The ToG method, although achieving decent recall, had the highest time cost. In contrast to GPT-4, Qwen2, and Llama3.1, Baichuan3 provided the fastest response times among all LLM-based methods, yet its recall performance was not particularly noteworthy. From a methodological perspective, Text2Cypher, KAPING, and KGQArr exhibited significantly low time costs; however, their recall was also quite limited, not exceeding 0.3.

Figure 6 (b) compares recall and token cost, with the horizontal axis again representing recall and the vertical axis representing token cost. Due to their minimal computational resource requirements and negligible financial costs, token costs are not

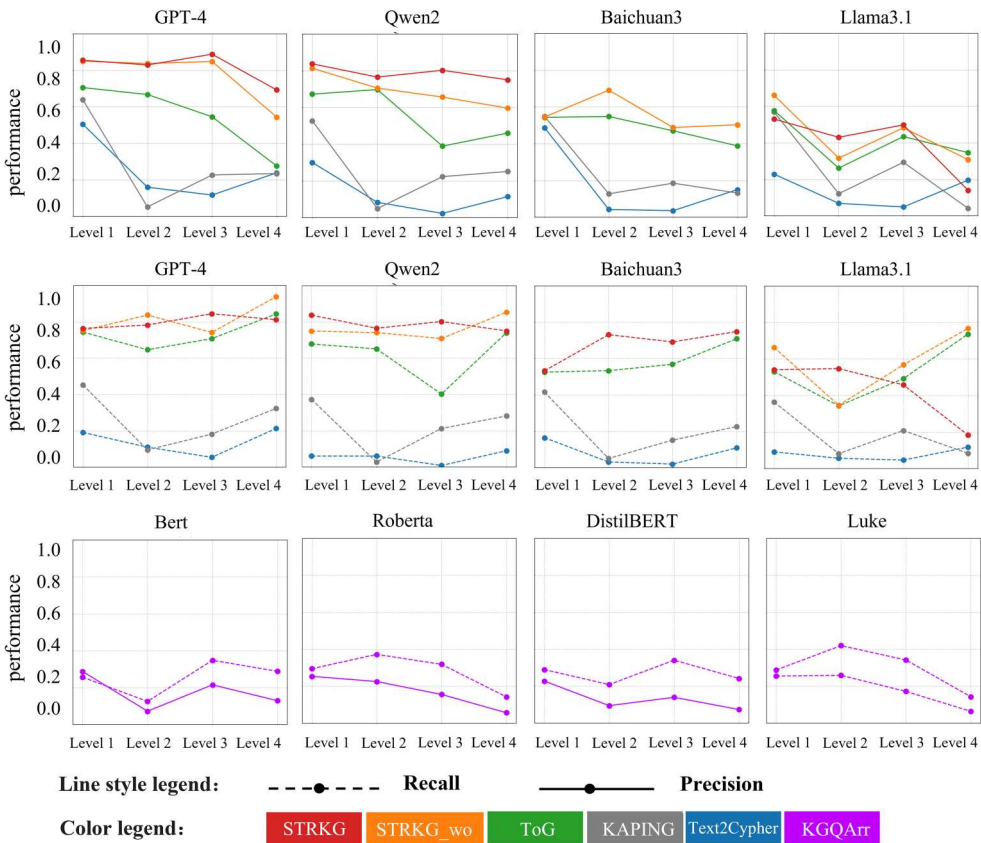


Figure 5. The precision and recall of competency questions at different levels.

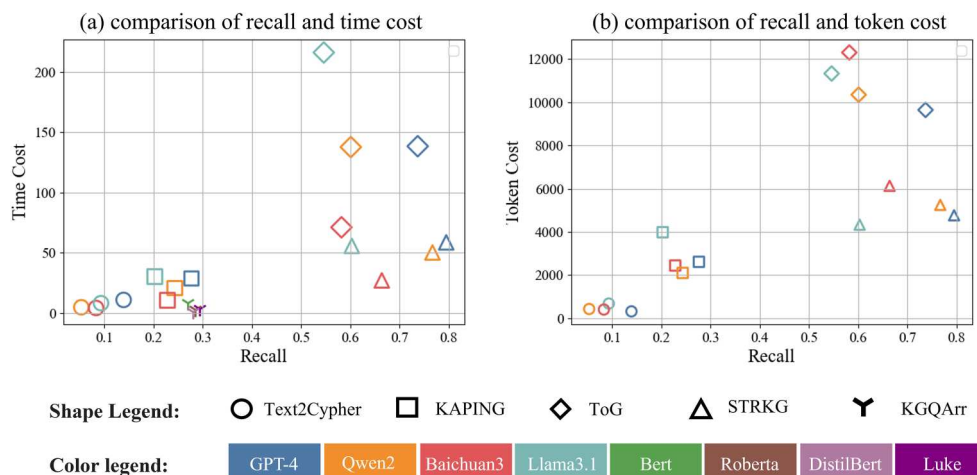


Figure 6. The recall, time cost, and token cost with different combinations of models and methods.

accounted for in methods based on PLMs. Compared to Text2Cypher and KAPING, we observed that STRKG and ToG displayed higher token costs but also achieved superior performance. Closer inspection reveals that the proposed STRKG method not only outperformed ToG in terms of performance but also significantly reduced token costs. In conclusion, analyzing time and token costs, along with recall, offers a comprehensive understanding of the efficiency and effectiveness of different methods and LLMs. This analysis emphasizes the need to balance these factors to optimize the overall performance of geospatial data question-answering systems. Future work should focus on enhancing methods to achieve high recall with lower resource consumption, thereby making these systems more accessible and practical for real-world applications.

4.5. Case elaboration and analysis

To clarify the STRKG method, this section outlines the reasoning process through two case studies.

The first case revolves around Sea-Surface Temperature (SST) data, which involve multiple constraints, multiple hops, and spatiotemporal conditions, as illustrated in Figure 7. First, ‘sea-surface temperature’ is identified as the user intent from the user’s query. Then, through semantic matching, this intent is linked to the nodes ‘SST’ and ‘Global gridded SST,’ which act as initial nodes. At depth 1, the LLMs determine that the ‘Global gridded SST’ node directly satisfies the user’s needs, thus concluding the exploration on this reasoning path. However, the ‘SST’ node does not fully meet the user’s query requirements, prompting STRKG to initiate further knowledge exploration along this node. In the subsequent node exploration and pruning, STRKG retains the nodes ‘Labrador Climate Index’ and ‘Monthly SST Product.’ However, the LLMs concluded that the current reasoning path does not sufficiently support answering the user’s question because it fails to match the critical spatial intent. STRKG then proceeds to the next step of exploration and reasoning. The next hop includes four nodes: ‘iceberg count,’ ‘Newfoundland,’ ‘global,’ and ‘disaster prevention.’ During the relation pruning

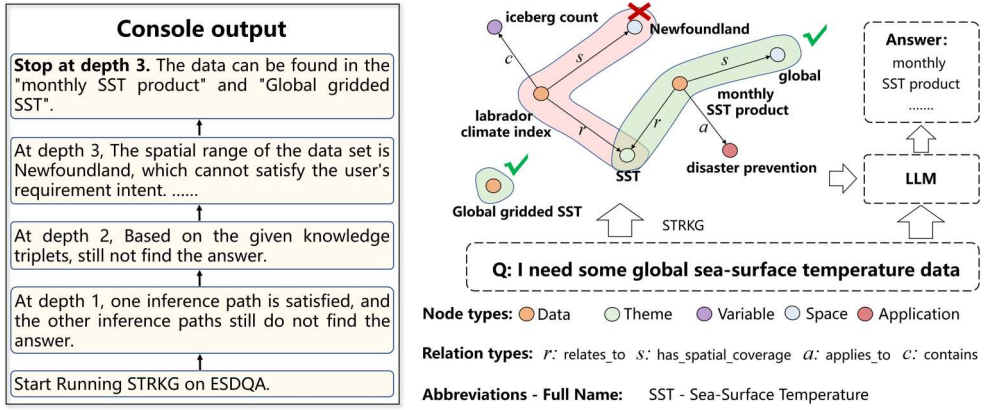


Figure 7. Detailed illustration of the reasoning process for the first case.

process, the LLMs discard the relation 'applies_to' since it does not address the user's query. The reasoning path subsequently reaches the 'Newfoundland' and 'global' nodes. Given the user's spatial intent for a global dataset, the simplified reasoning path '(Node: Theme, SST) - (Relation: relates_to) -> (Node: Data, monthly SST product) - (Relation: has_spatial_coverage) -> (Node: Space, global)' is deemed to contain the correct information.

The second case focuses on precipitation and incorporates multiple constraints, spatiotemporal conditions, and inference conditions, as illustrated in Figure 8. Initially, the node 'Hydrological Global Forcing Data' is not identified directly through semantic matching. One of the initial nodes of the reasoning path is 'daily precipitation,' derived from unstructured text via the information extraction model. In subsequent exploration stages, STRKG identifies the target data 'Hydrological Global Forcing Data' through the reasoning path '(Node: Theme, daily precipitation) - (Relation: has the variable of) -> (Node: Data, Hydrological Global Forcing Data).' This demonstrates the

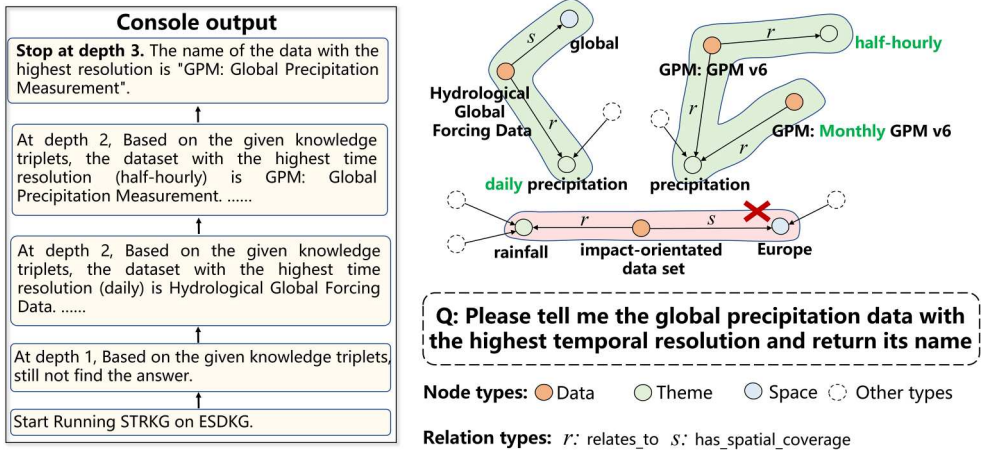


Figure 8. Detailed illustration of the reasoning process for the second case.

significant role of mining unstructured text, which aids STRKG in finding one of the correct answers through the reasoning path. Meanwhile, another reasoning path '(Node: Theme, rainfall) – (Relation: relates_to) – > (Node: Data, impact-orientated data set) – (Relation: has_spatial_coverage) – > (Node: Space, Europe)' is directly pruned because the dataset's spatial scope does not match the spatial intent of the user's query. Given the complexity of the question, STRKG does not easily determine the final answer directly through node exploration. Instead, STRKG maintains multiple potential reasoning paths and utilizes the LLMs' multiple follow-up inquiries to ultimately match the answer. In this example, by exploring three depths, STRKG successfully identifies the correct answer. This success is attributed to the ongoing maintenance of multiple reasoning paths and the ability to conduct knowledge reasoning from a broader perspective. However, there are still some shortcomings with GDQA, which we discuss in detail in the discussion section.

5. Discussion

The GDQA system represents a novel technological paradigm for geospatial data retrieval tasks by integrating knowledge graphs with LLMs. This integration provides significant advantages over traditional keyword-matching techniques or standalone knowledge graph technologies. Specifically, this approach substantially lowers the barrier for users in geospatial data retrieval and discovery. However, despite its promising potential, two critical challenges need to be addressed.

One challenge pertains to resource optimization. While GDQA demonstrates strong performance across a range of competency questions, its high demand for computational resources is a significant concern. The system relies heavily on multiple calls to LLMs for knowledge reasoning and pruning on the GDKG. For complex queries involving multiple constraints and multiple hops, the computational load can increase exponentially, leading to higher costs and longer response times. Additionally, this process accelerates the consumption of the LLMs' limited token resources. As the knowledge graph continues to expand, the rate of token consumption is expected to increase further. Therefore, optimizing the processes for knowledge exploration and pruning to manage token resource consumption will be a critical focus of future research. Techniques such as efficient sampling methods, caching frequently accessed data, and developing lightweight reasoning algorithms could help mitigate these resource demands.

The second challenge concerns performance. The performance of the question-answering system largely depends on two aspects: (1) the accuracy of user intent recognition, and (2) the intelligence of the LLMs. In terms of intent recognition, the current approach in our work uses LLMs to extract entities from the user's question to determine their intent. However, in the highly complex field of geographic information, solely relying on entity extraction to determine user intent may overlook implicit intents. For example, in the question 'I want to obtain some remote sensing imagery data of the main spring wheat production areas in China', the phrase 'main spring wheat production areas in China' implicitly contains the user's spatiotemporal intent, roughly referring to the northeastern region of China. Traditional intent recognition methods are unlikely to capture such implicit intents. One potential direction for addressing this challenge is to develop LLMs specifically for Earth sciences. This could involve

pre-training on vast amounts of Earth science text data and fine-tuning using instruction data from the geoscience domain. Such efforts would provide the LLMs with deeper domain knowledge, enhancing their ability to recognize user spatiotemporal intent and perform spatiotemporal reasoning.

In summary, while GDQA represents a promising advancement in geospatial data retrieval, it is essential to address challenges related to resource optimization, performance, scalability, and user experience for its successful deployment and adoption in real-world applications. By focusing on these areas, we can unlock the full potential of GDQA systems and facilitate more efficient and accessible geospatial data discovery.

6. Conclusions

In this study, we introduced a generative question-answering system enhanced by a geospatial data knowledge graph, named GDQA. Initially, we constructed a geospatial data knowledge graph, named GDKG, which aggregates geospatial data from five different data portals. GDKG incorporates fine-grained knowledge extracted from abstracts of the academic literature, creating annotations and links between various resources. We then proposed a spatiotemporal knowledge reasoning method based on LLMs, named STRKG, which effectively leverages the strengths of both the knowledge graph and LLMs. GDQA enables both professional and non-professional users to retrieve and discover geological data using natural language. To assess the effectiveness of our approach, we compared it against conventional methods such as Text2Cypher and ToG, using a set of competency questions. The experimental results demonstrate that our method outperforms the others in several challenging dimensions, including multiple constraints, multiple hops, spatiotemporal conditions, and inference conditions. Additionally, we provided detailed case studies that illustrate how STRKG generates and explores reasoning paths step by step. Ultimately, GDQA provides data query and discovery services to users within the Earth science community.

Notes

1. Links: <https://www.earth-system-science-data.net/>
2. Links: <https://developers.google.cn/earth-engine/datasets/catalog>
3. Links: <https://cds.climate.copernicus.eu>
4. Links: <https://gee-community-catalog.org/>
5. Links: <https://www.resdc.cn/>
6. Links: <https://doi.org/10.57760/sciencedb.10436>
7. Links: <https://huggingface.co/facebook/bart-large>
8. Links: <https://platform.openai.com/docs/models/gpt-3-5-turbo>
9. Links: <https://huggingface.co/BAAI/bge-m3>
10. Links: <https://platform.openai.com/docs/models/gpt-4-turbo-and-gpt-4>
11. Links: <https://bailian.console.aliyun.com/>
12. Links: <https://platform.baichuan-ai.com/>
13. Links: <https://huggingface.co/meta-llama/Llama-3.1-70B-Instruct>

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The work was supported by National Natural Science Foundation of China (No. 42425108), Chongqing Technology Innovation and Application Development Project (No. CSTB2024NSCQ-LZX0148), Nature Science Foundation Innovation Group Project of Hubei Province, China (No. 2025AFA035), Hubei LuoJia Laboratory, State Key Laboratory of Intelligent Vehicle Safety Technology, and Chongqing Changan Automobile Co. Ltd.

Data and codes availability statement

The data and codes supporting the findings of this study are available at <https://github.com/openrsgis/GDQA>.

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